

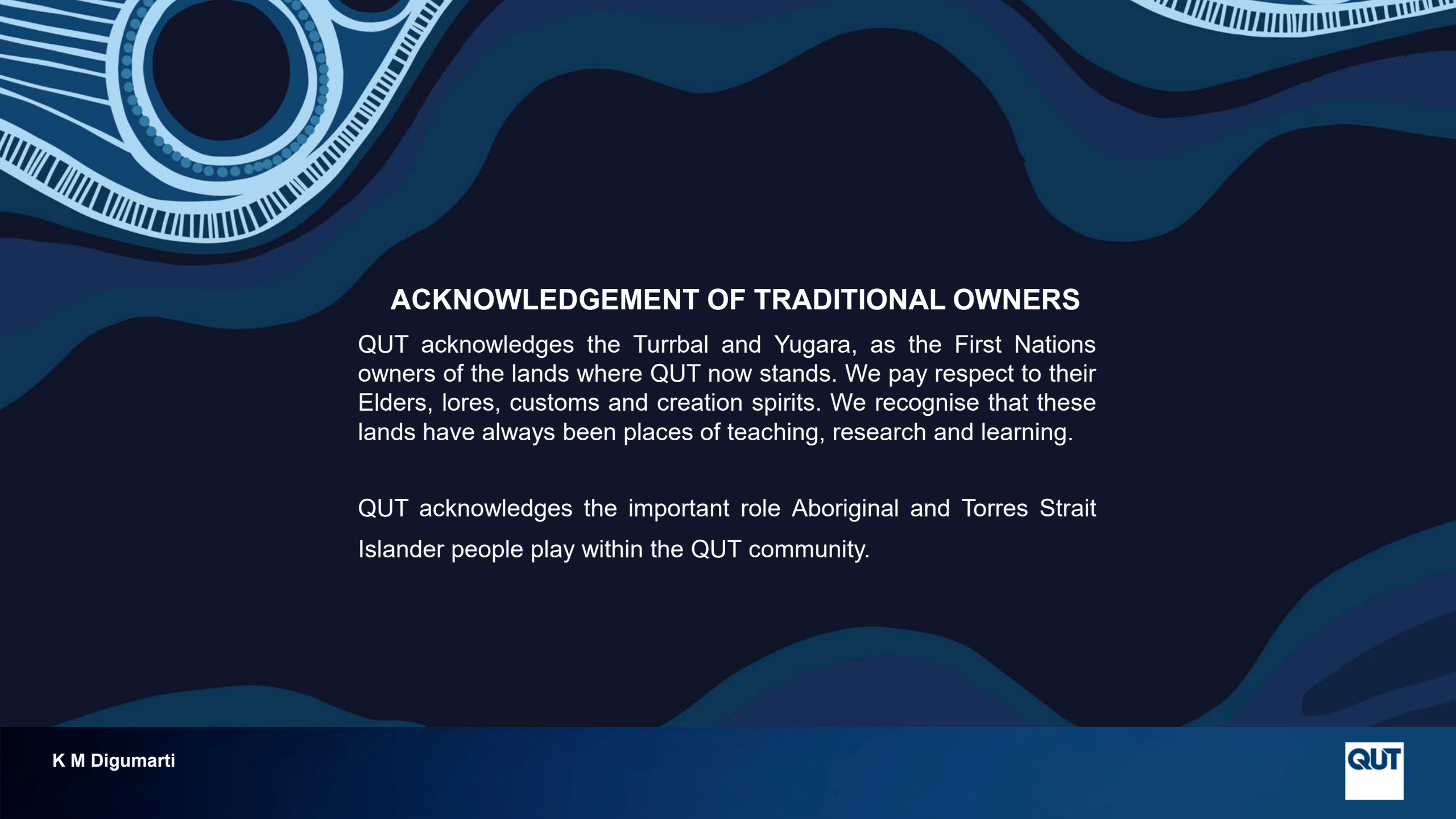


Building a shockingly good tentacle

Krishna Manaswi Digumarti

Lecturer, QUT Centre for Robotics

04 July 2025

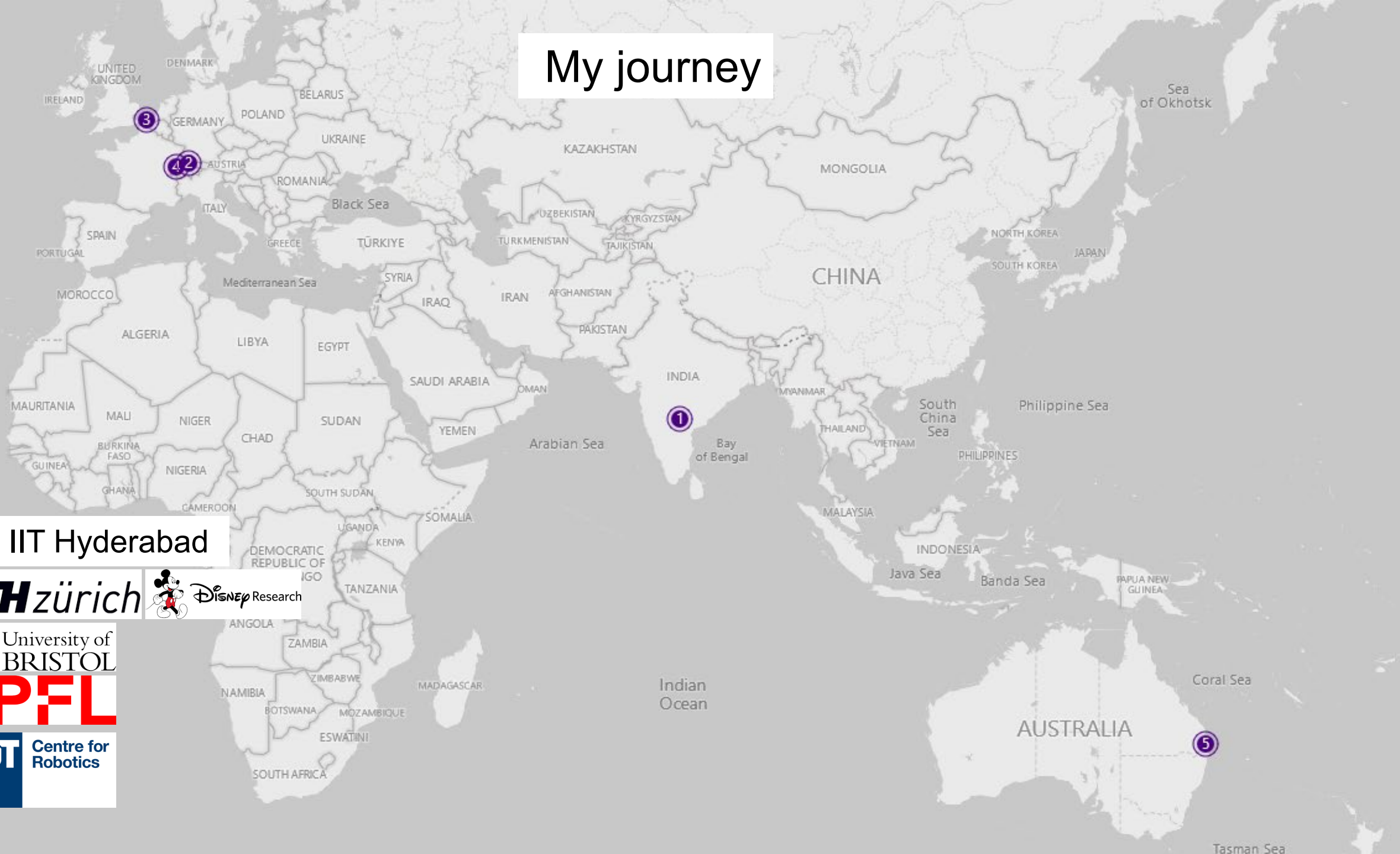


ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

My journey



1



IIT Hyderabad

2

ETH zürich



3

University of
BRISTOL

4

EPFL

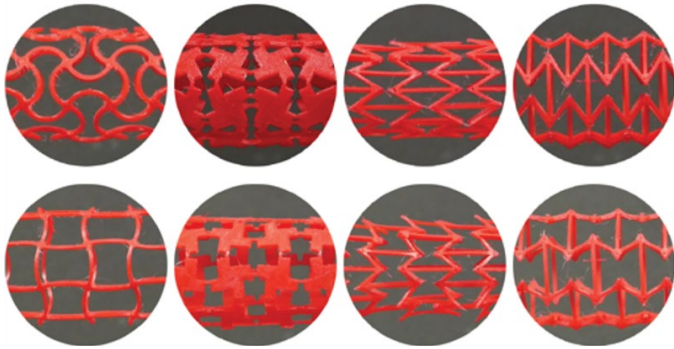
5

QUT
Centre for
Robotics

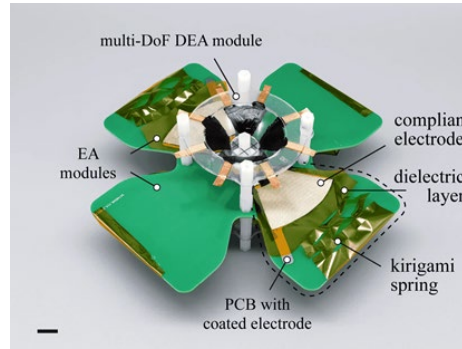
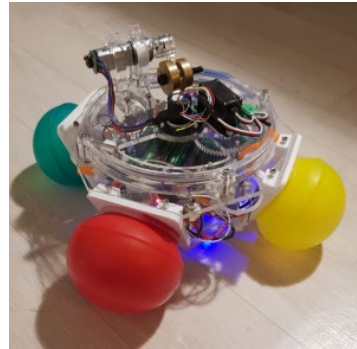
My interest is in soft robotics



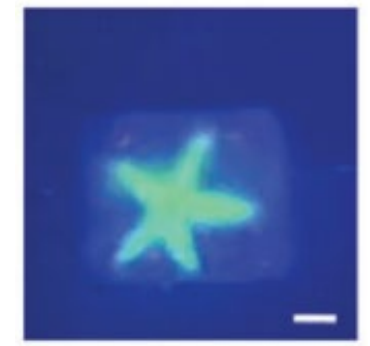
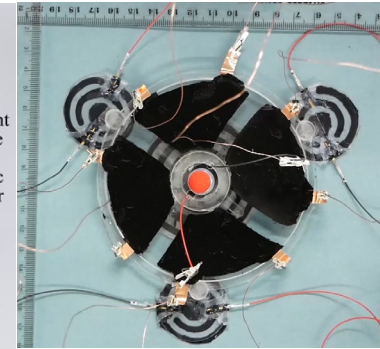
I build soft devices



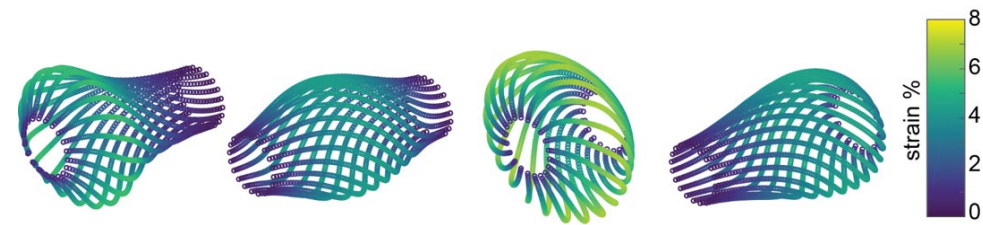
Tunable metamaterials



Robot locomotion



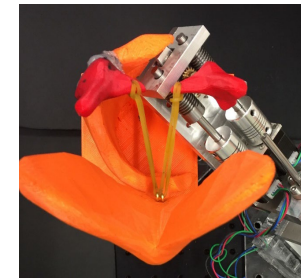
3D printing



Bio-inspired shape change

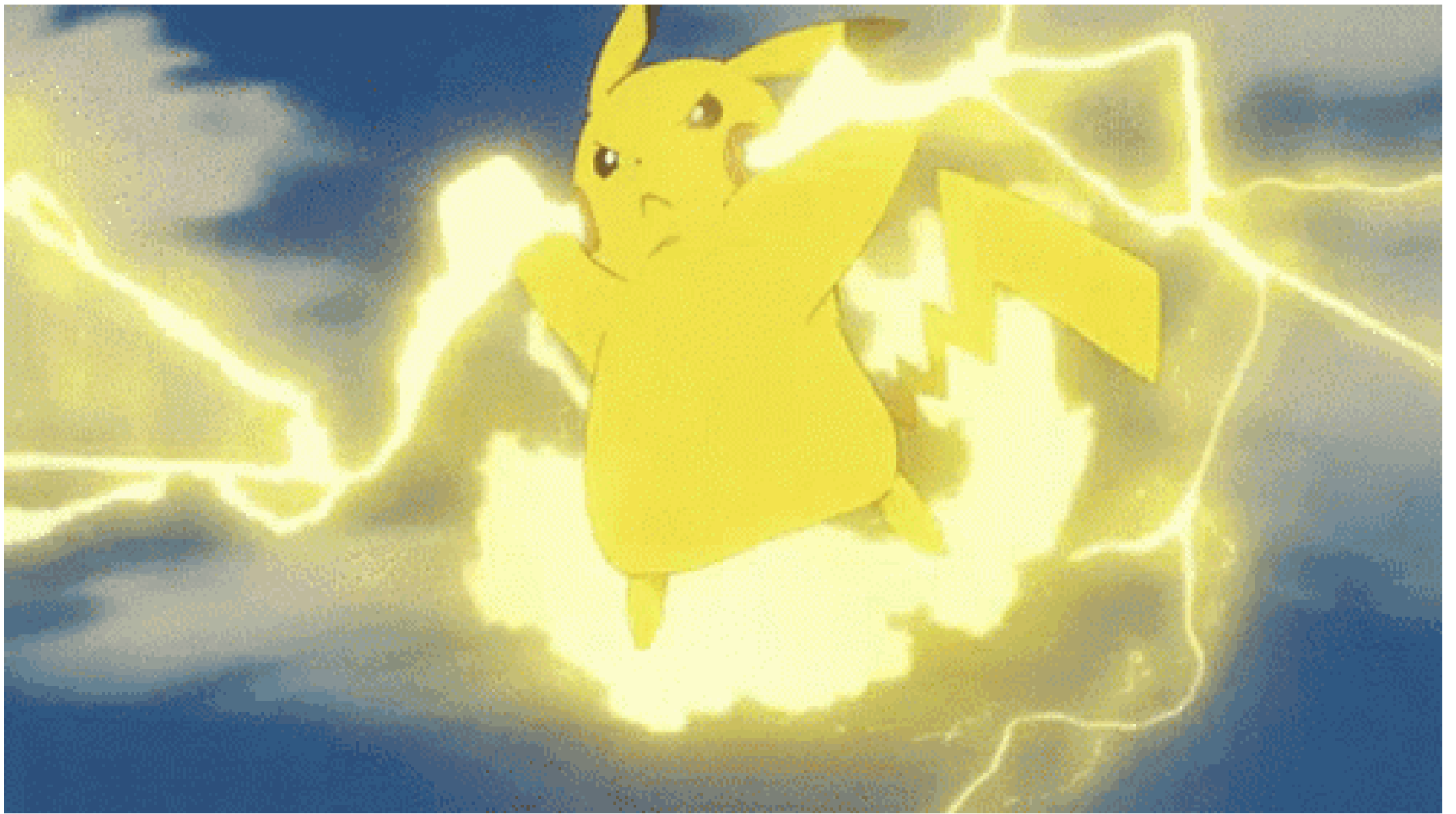


Tactile sensors



Medical devices





Exploiting electrostatic phenomenon for robotics applications

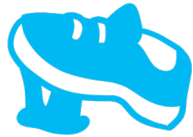
With a focus on soft polymer devices



... and a deeper dive into tactile sensing



Stiffness change



Adhesion



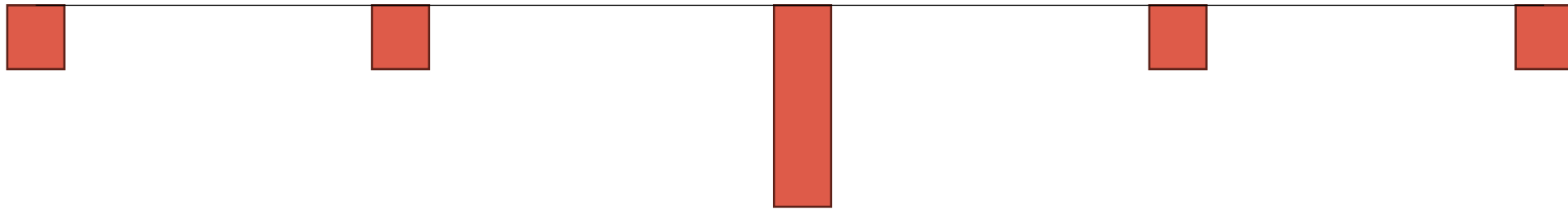
Sensing



Actuation



Haptics



The ability to change stiffness is useful in soft manipulators

- A soft state in which the manipulator can readily deform or wrap around objects
- A rigid state in which the manipulator exerts force

Electrically induced stiffness change

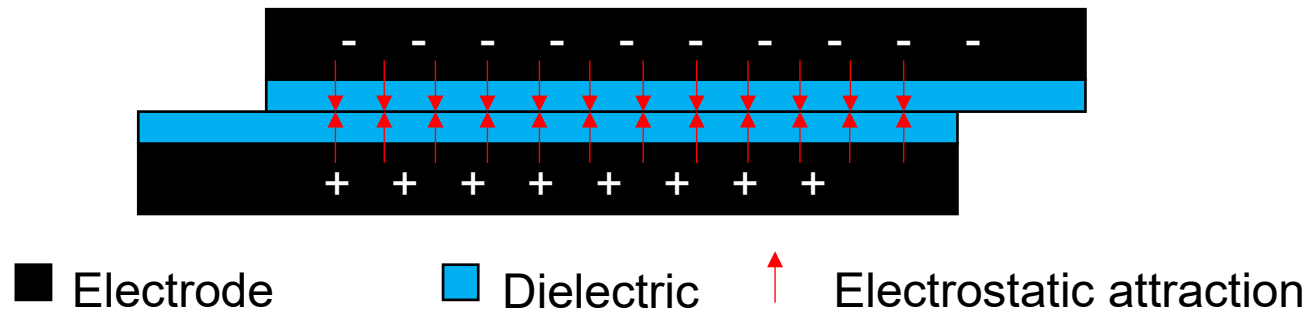
Jamming is one way to change stiffness

Interaction between neighbouring constituent elements allows a deformable structure to transition between states of low and high stiffness



Brown et al., PNAS, 2010

We can use electrostatic attraction to jam layers



Advantages of electroadhesive jamming

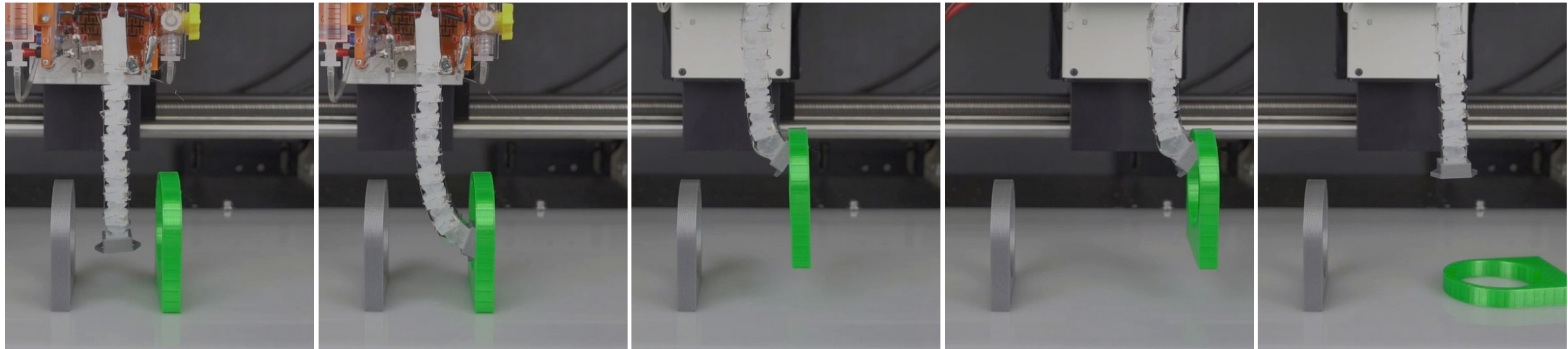
- Attractive pressures up to (10-15 N/cm²)
- Minimal change in volume¹
- Fast actuation (milliseconds for electrostatics compared to 10s of seconds for other methods)²
- External infrastructure is small (10⁻³ m³)
- Ability to regulate stiffness (not just two extremes)^{2,3}
- Possible to sense relative motion between electrodes
- Low power consumption (mW)

1. Fitzgerald et al., *Actuators*, 2020

2. Hinchet et al., *Adv Mat Tech*, 2020

3. Ramachandran et al., *Adv Mat Tech*, 2018

Jamming combined with an actuator allows directed deformation in addition to stiffness change

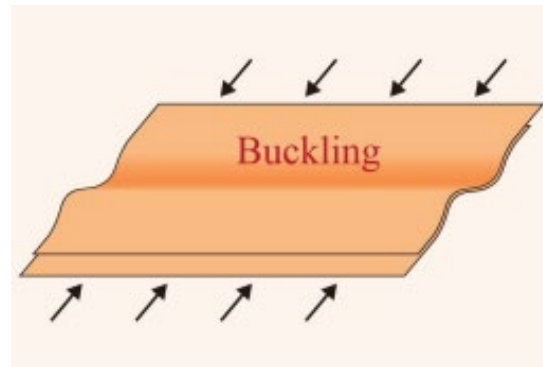
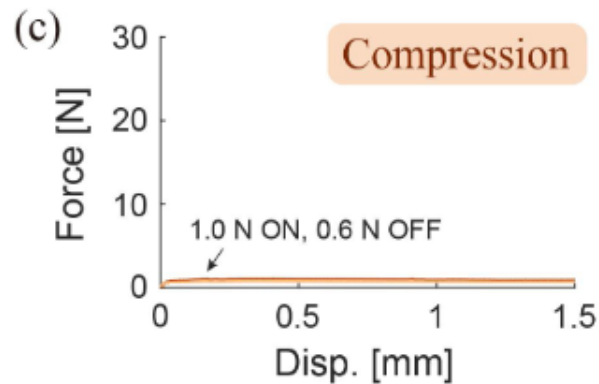
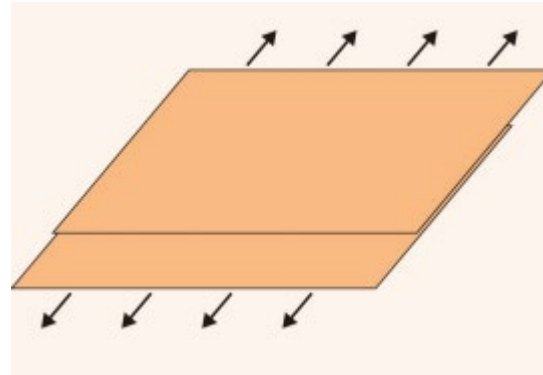
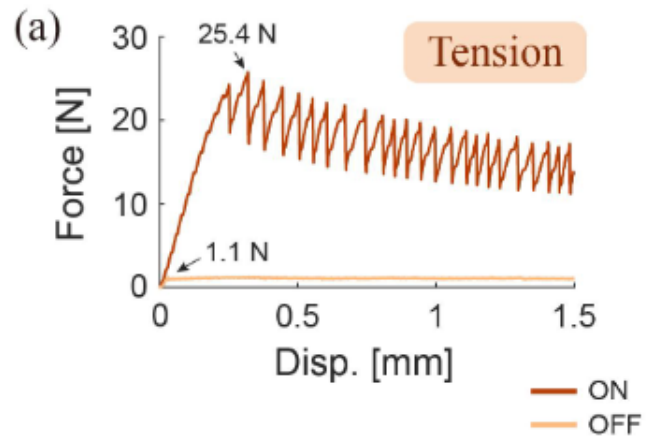


2 cm

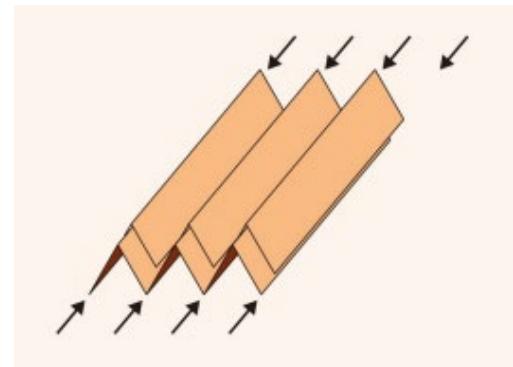
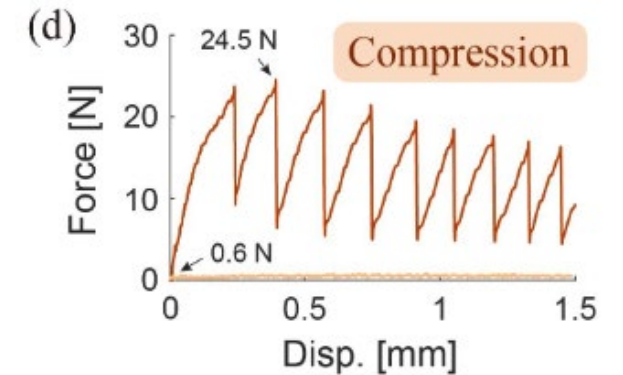
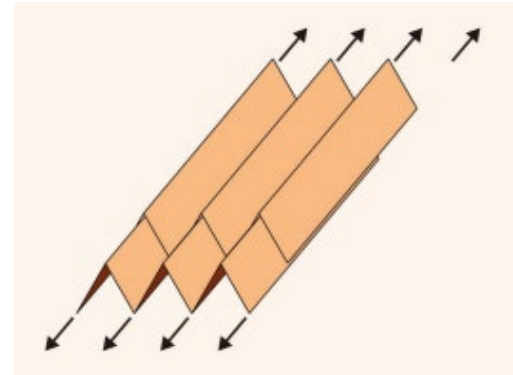
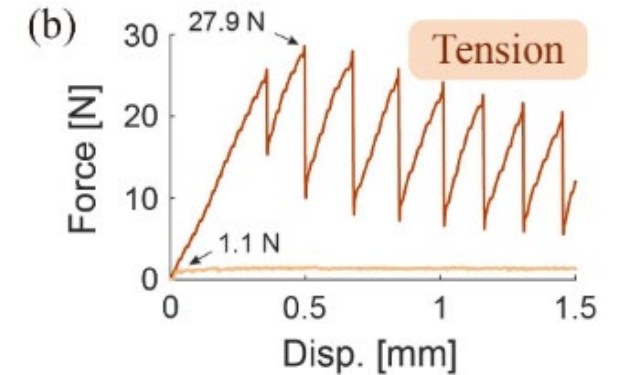
Up to 7 times change in flexural rigidity

Adding corrugations improves stiffness

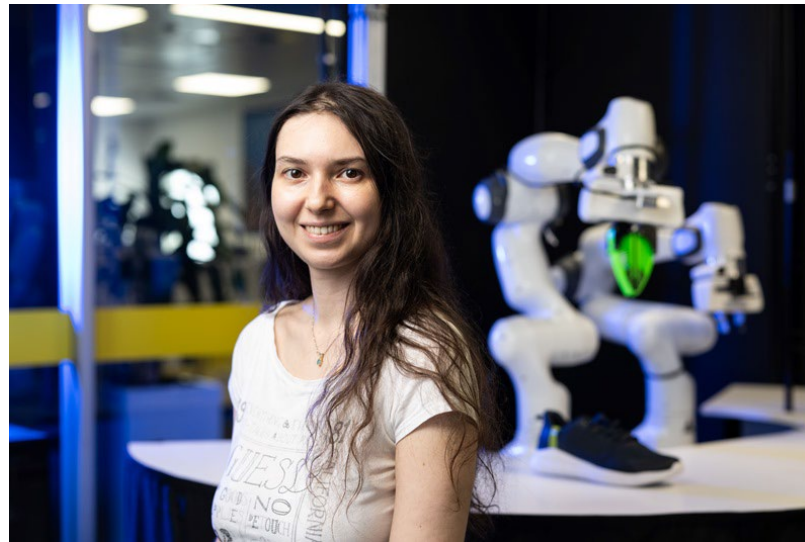
Planar EA clutch



Corrugated EA clutch



How do we integrate these structures in robot arms, fingers or tentacles?

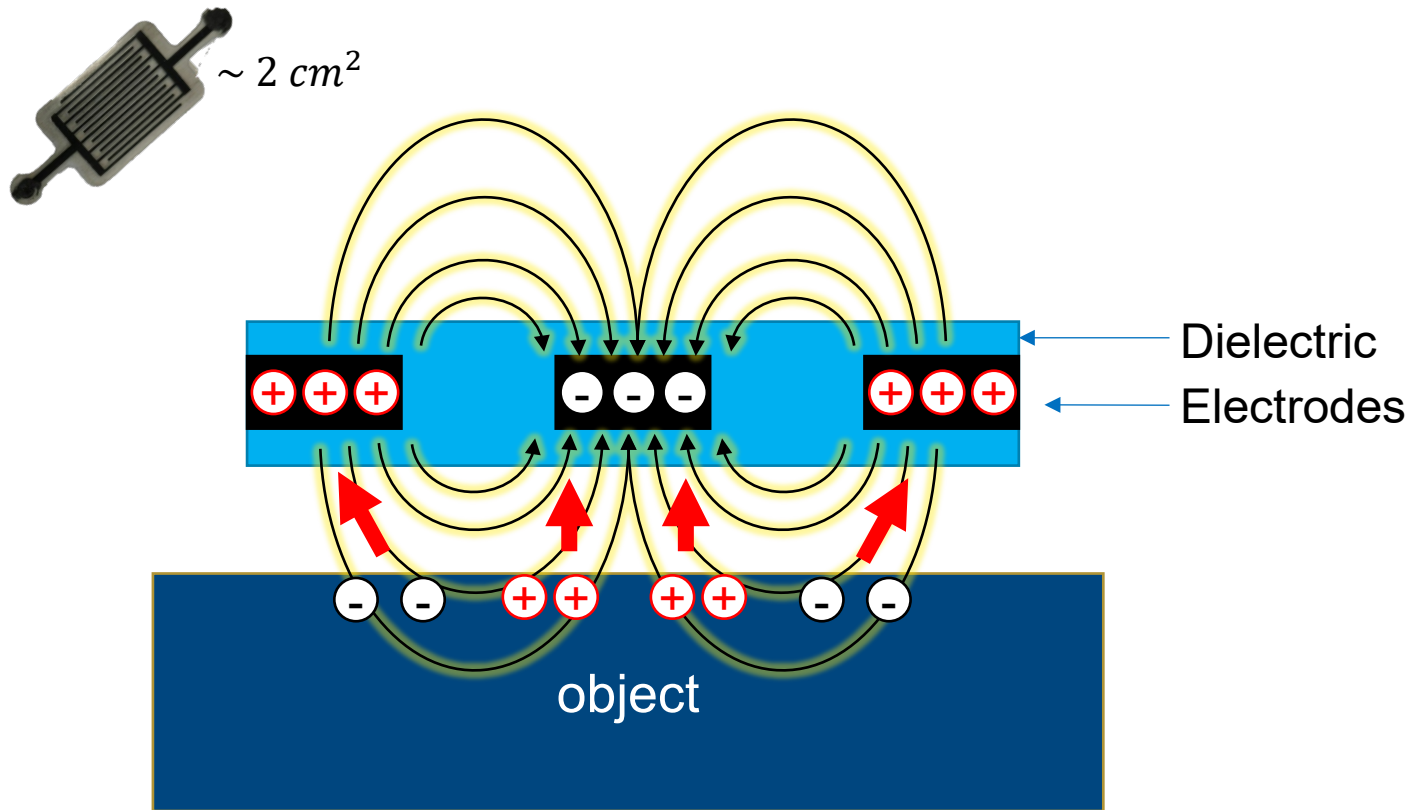


Jessica Gumowski

Stiffness and Shape Modulation for Multi-functional Amphibious Soft Robots

Adhesion for grasping

Electro-adhesion



Force of attraction

$$F \propto \epsilon_r A \frac{V^2}{d^2}$$

Relative permittivity

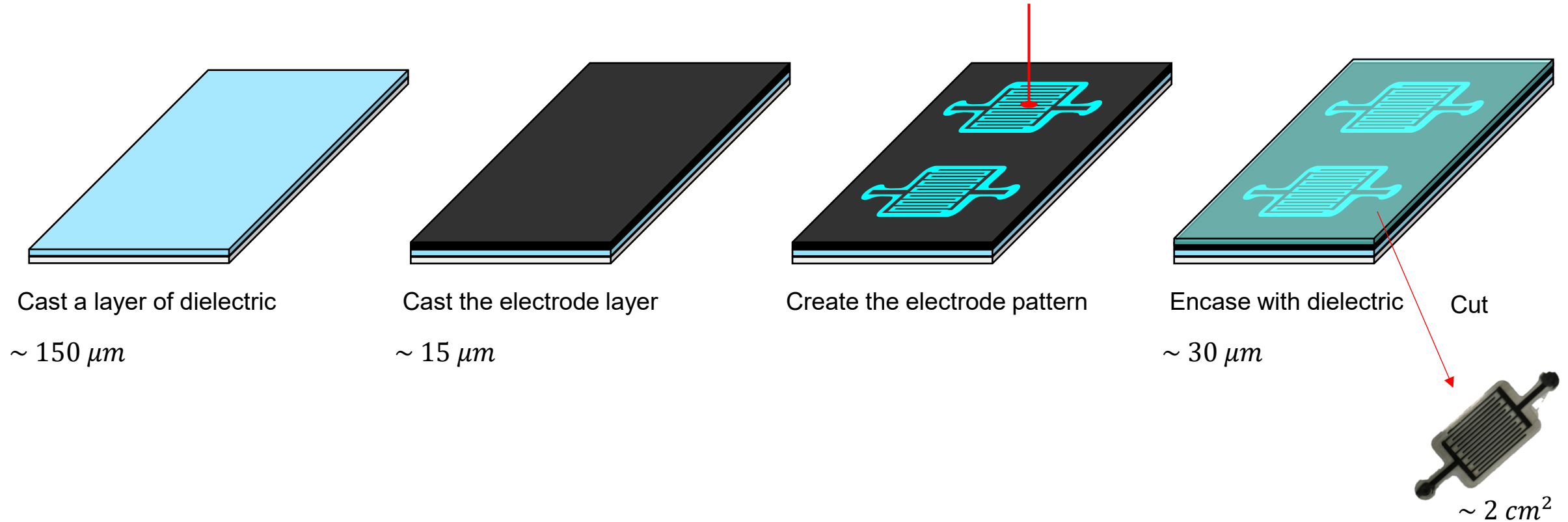
Area of overlap

Voltage

Separation

The field strength is usually kV/mm.

Electroadhesive devices are easy to fabricate



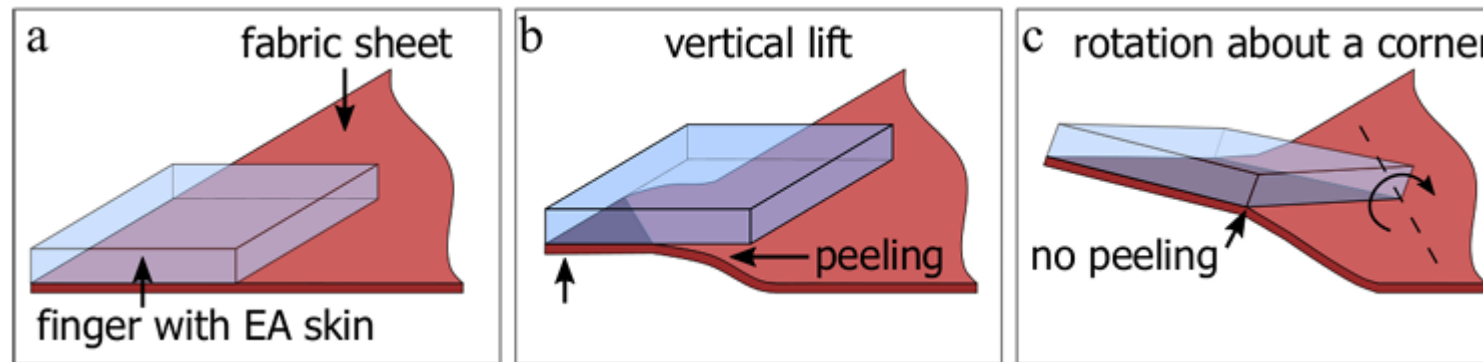
Picking something up from a stack is a common task



Selmark – a garment manufacturer in Spain

Electroadhesion is more nuanced than just touching and lifting – depends on the object and gripper

In practice, picking things up is more nuanced





In this use case,
the partner Omnigrasp



Benefits of electroadhesion

- Sticking and releasing is quick – takes milliseconds
- Works for both conductive and insulating materials
- Does not damage the object – e.g. through pinching or piercing
- Works with objects of various sizes and shapes
- Force of attraction is highest at the interaction surface
- Enables non-prehensile grasps

How do we know if we picked anything up?

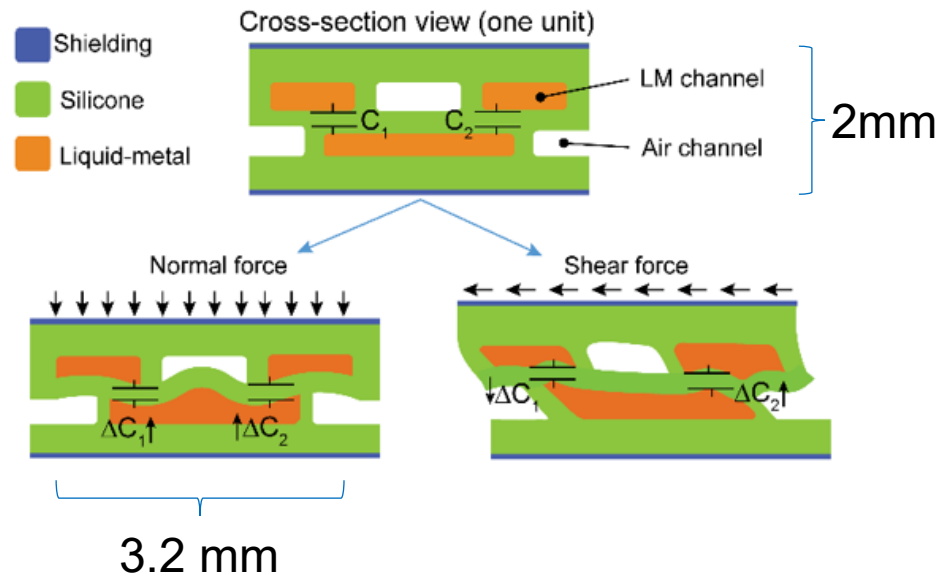
Are we holding the object firmly?

Is the object slipping?

We need a force sensor

Here is our force sensor

This sensor has tiny capacitors inside



Absolute permittivity

Area

Capacitance

$$C = \frac{\epsilon A}{d}$$

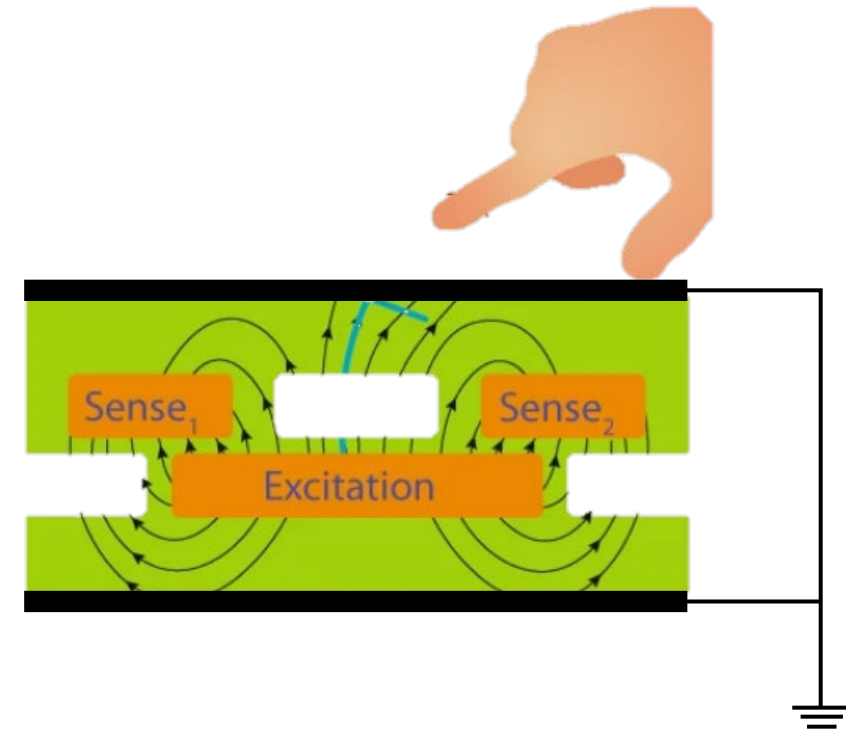
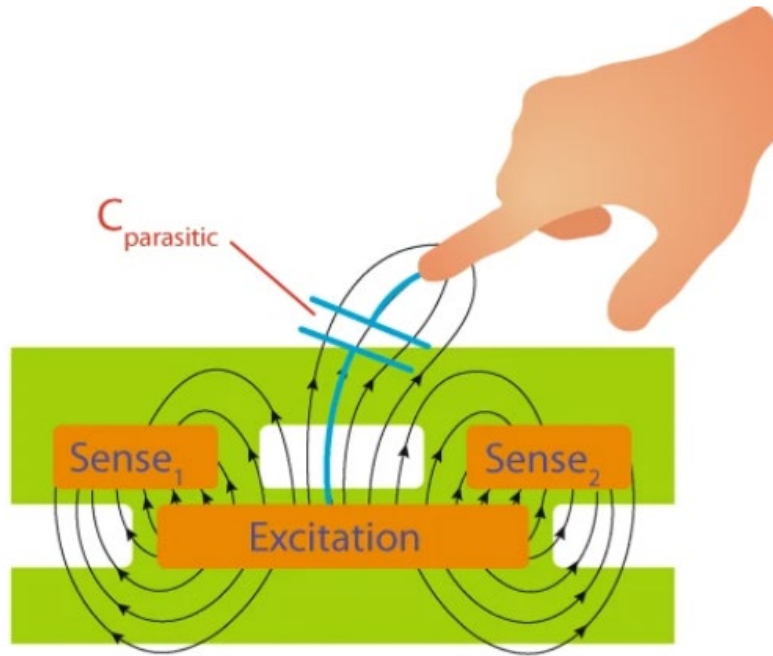
Distance b/w electrodes

Electrodes are made of liquid metal (E Galn)
- compliant and maintain conductivity when deformed

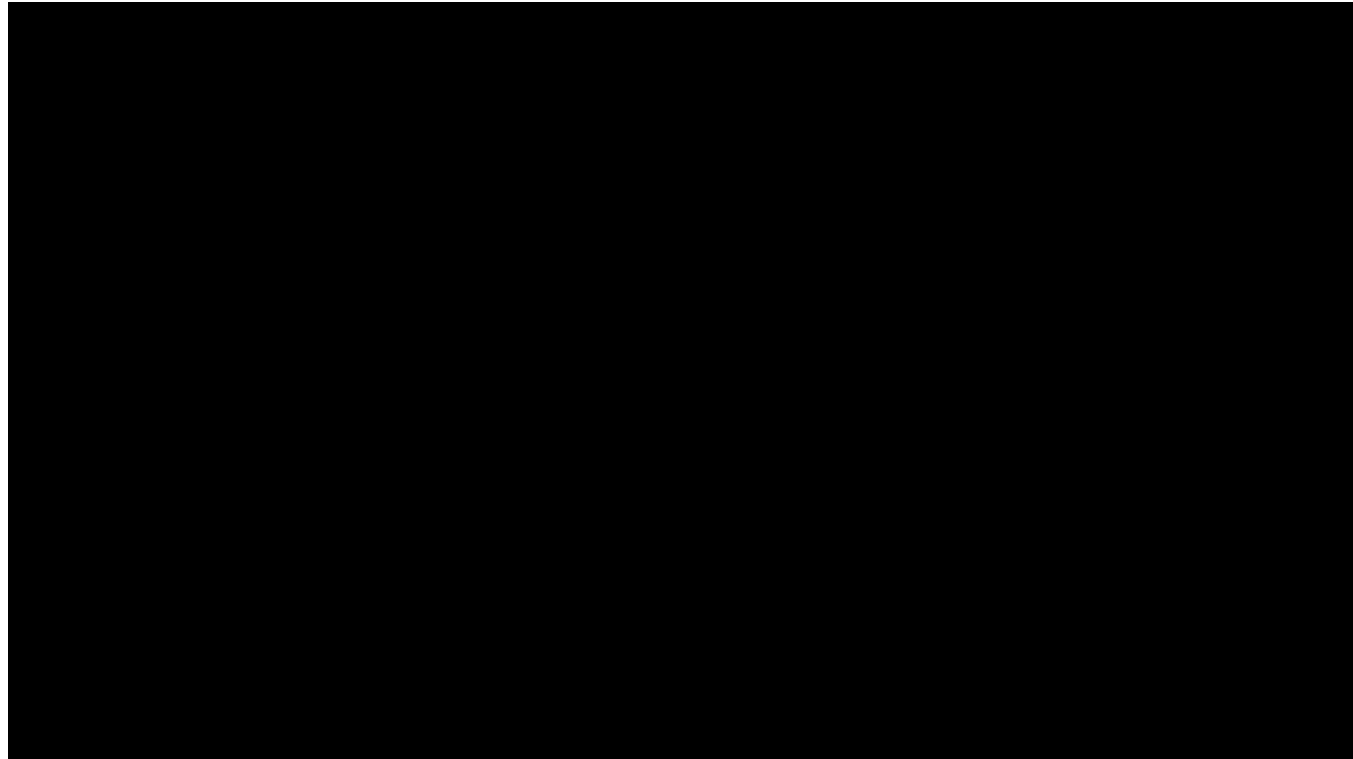


Capacitive sensors are susceptible to noise

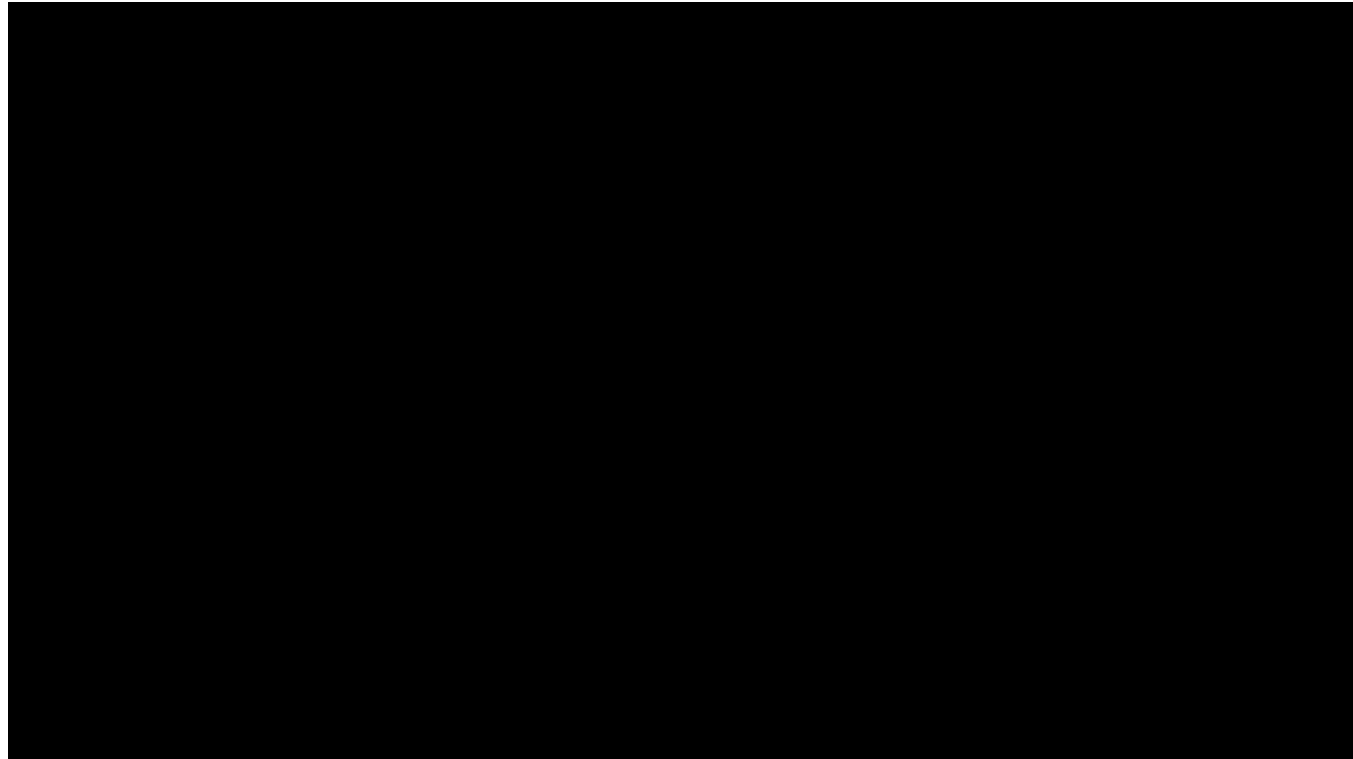
Shield the capacitors from external fields



Shielding works well

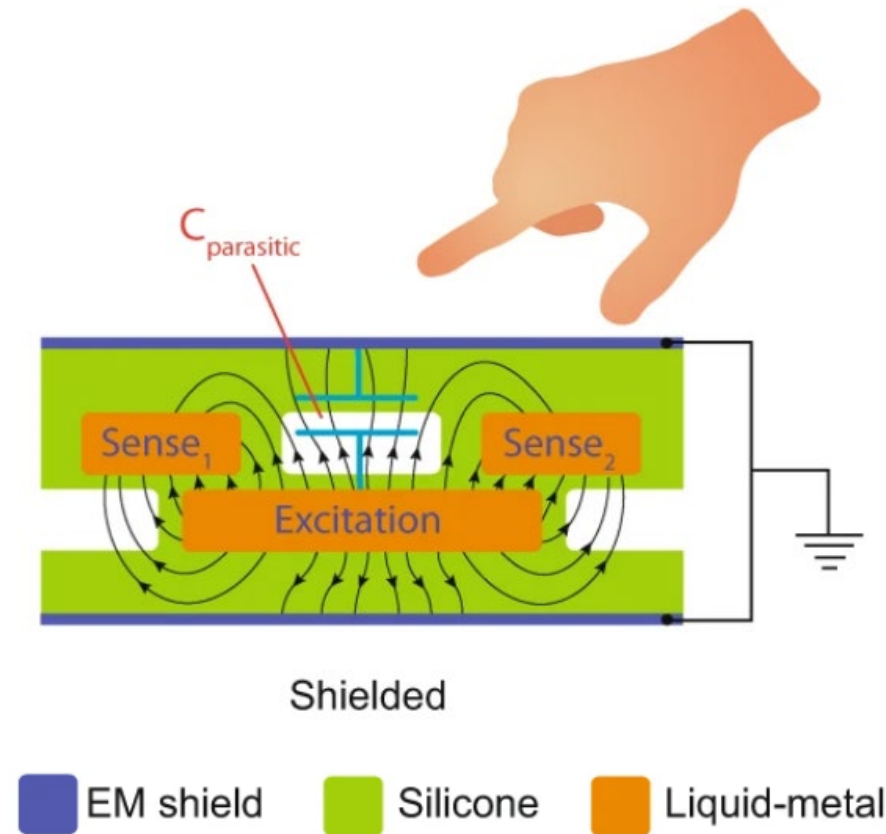


The sensor is protected from external electric fields

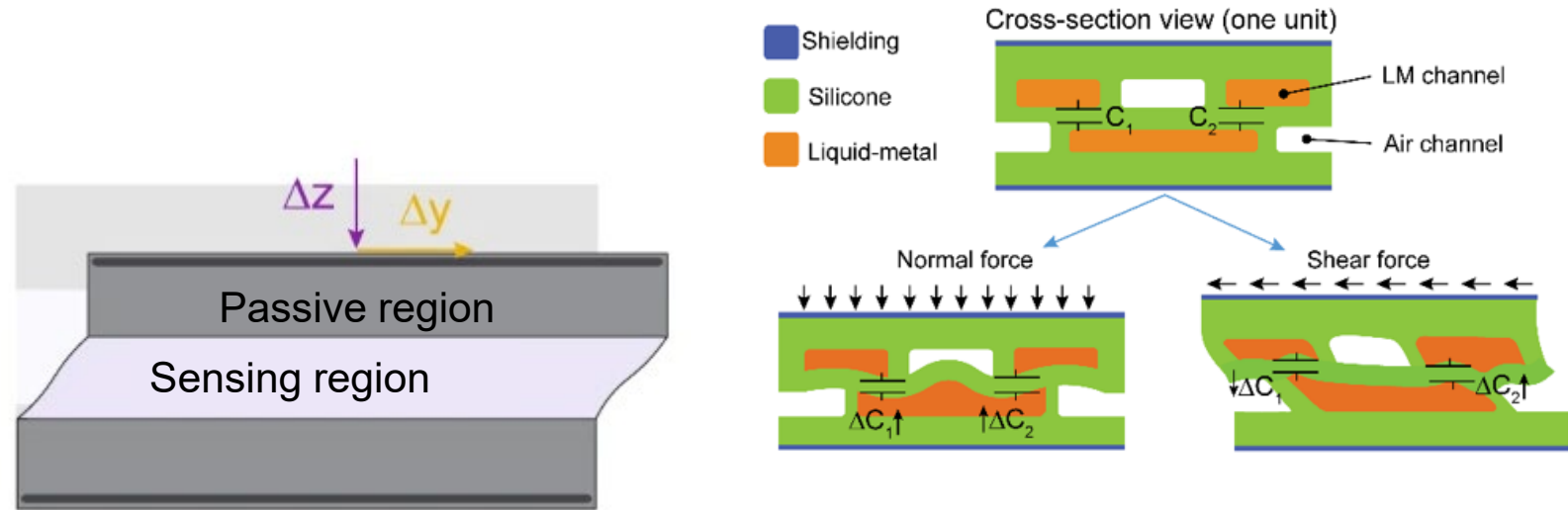


But, shielding introduces a new problem

Introduces parasitic capacitance and reduces sensitivity



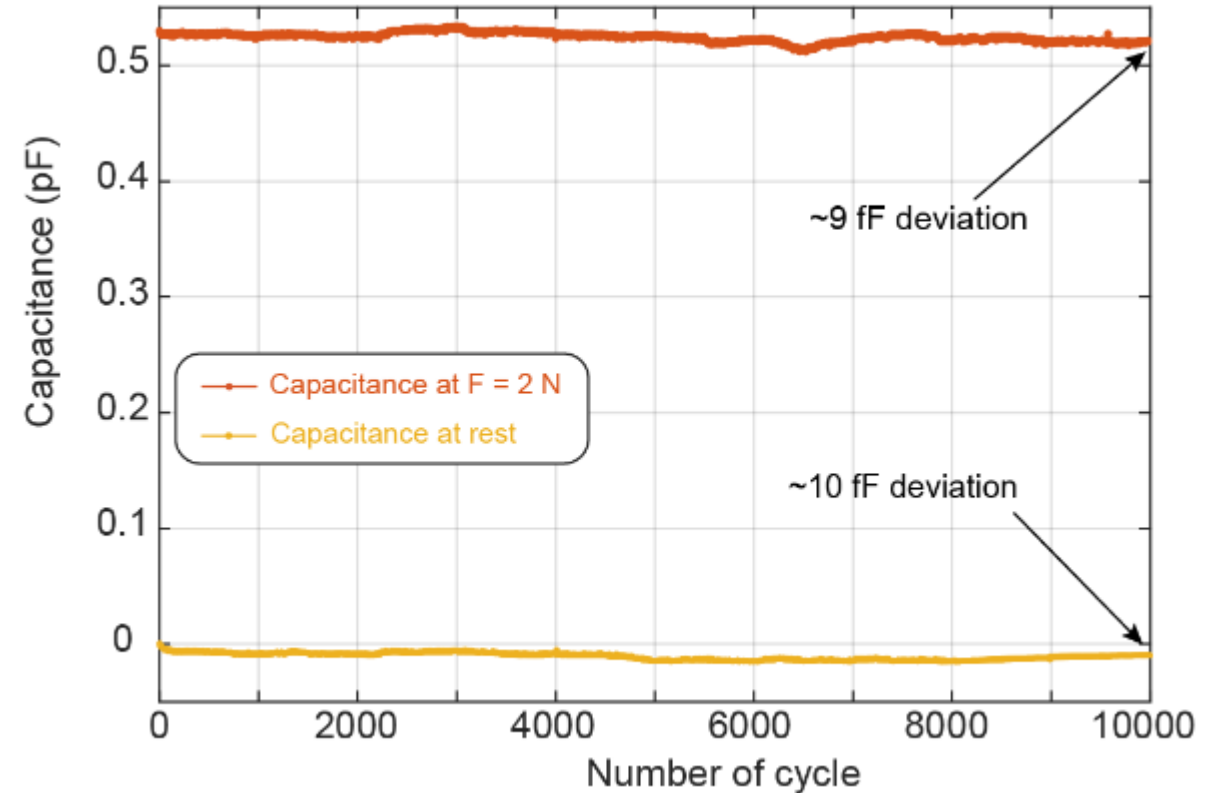
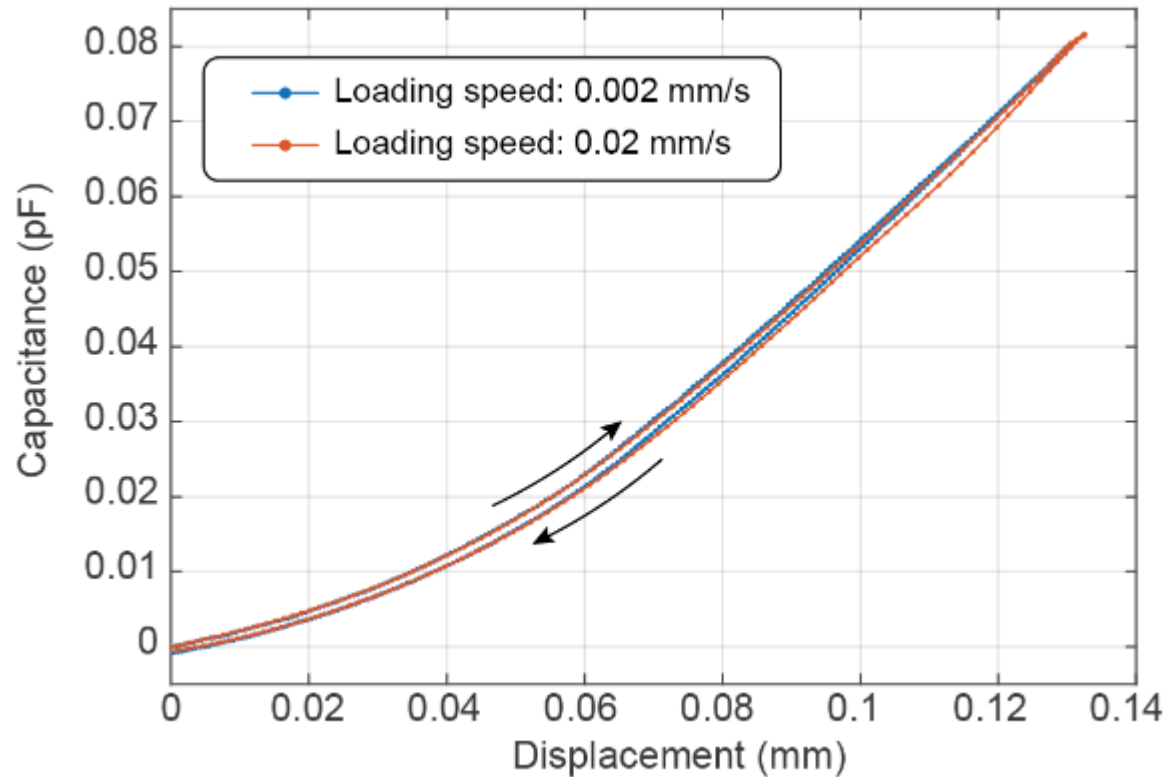
We want the capacitance to change only in the sensing region



Solution: Tune the mechanical stiffness of layers to make the sensing region relatively less stiff
The pattern of air channels and electrodes serves this purpose. Elastic modulus of material is also changed.

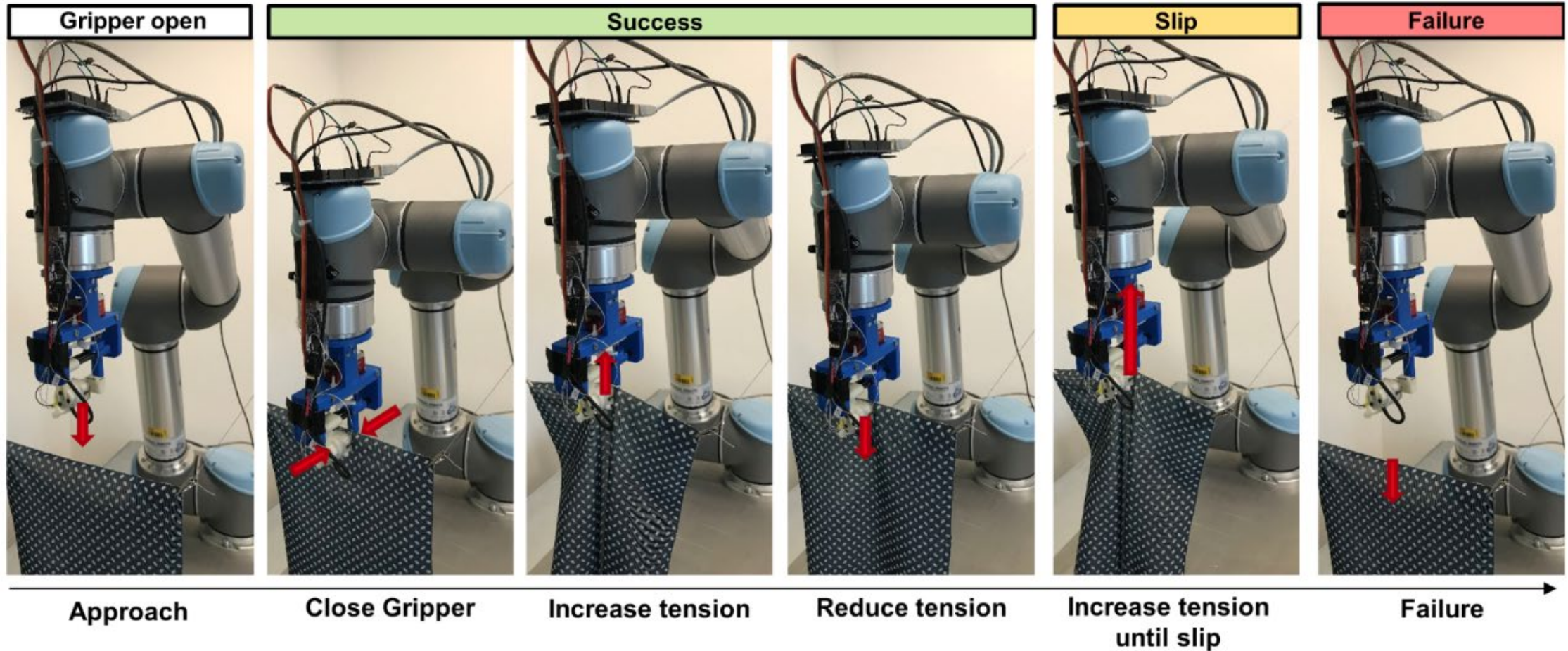
Relative deformation in the sensing region is 30 times higher than in the passive region

The sensor shows about 2% hysteresis and <35 mN deviation after 10,000 cycles of loading



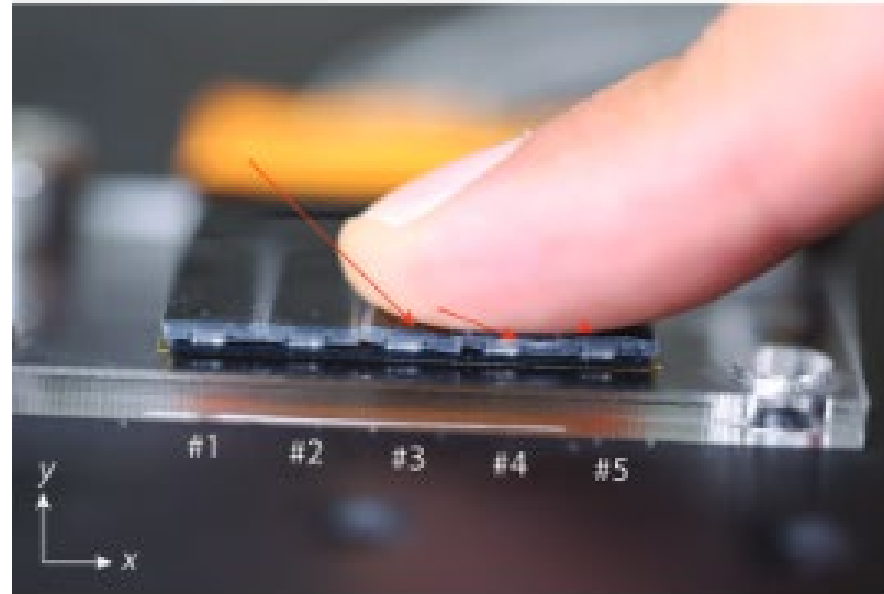
Plenty of applications for force sensors

We used the sensor to detect slip



One sensor is nice, can we have more?

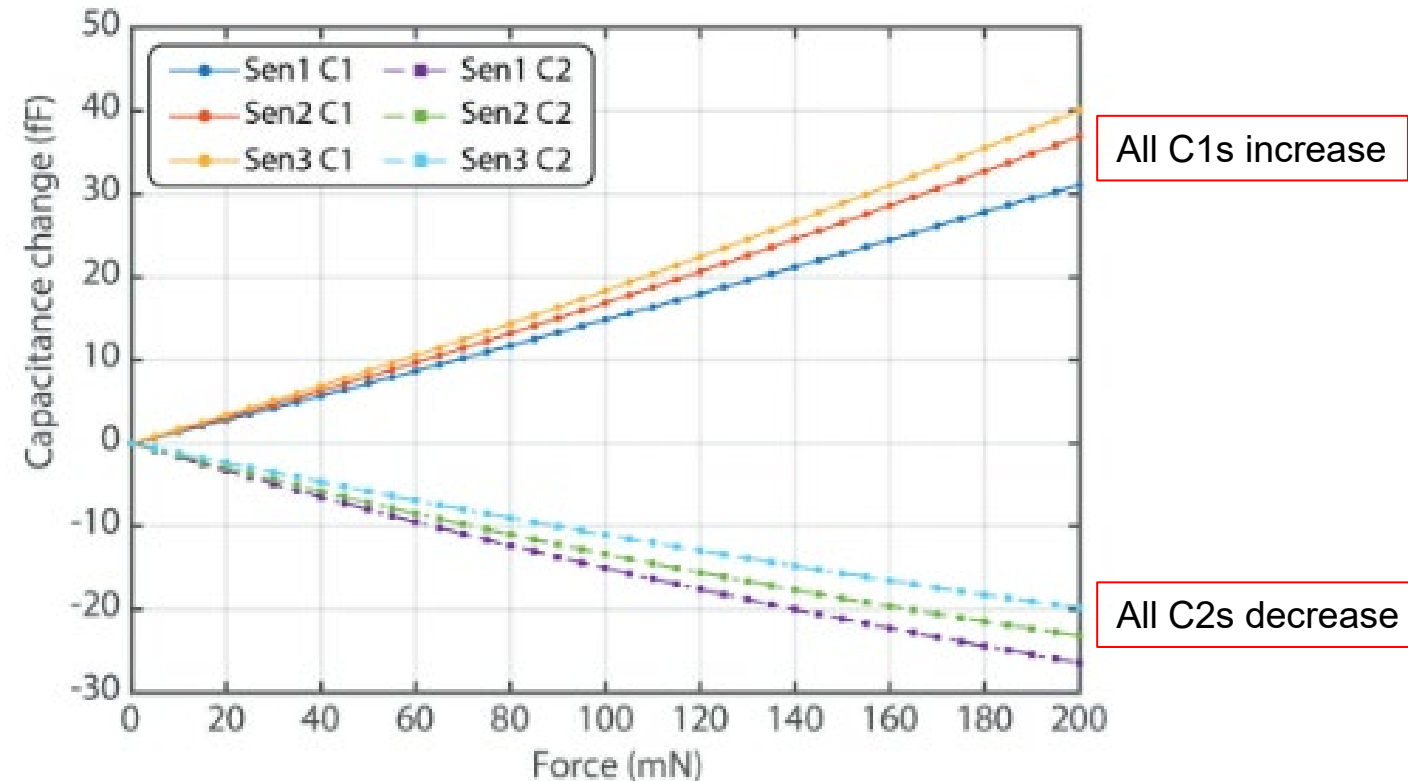
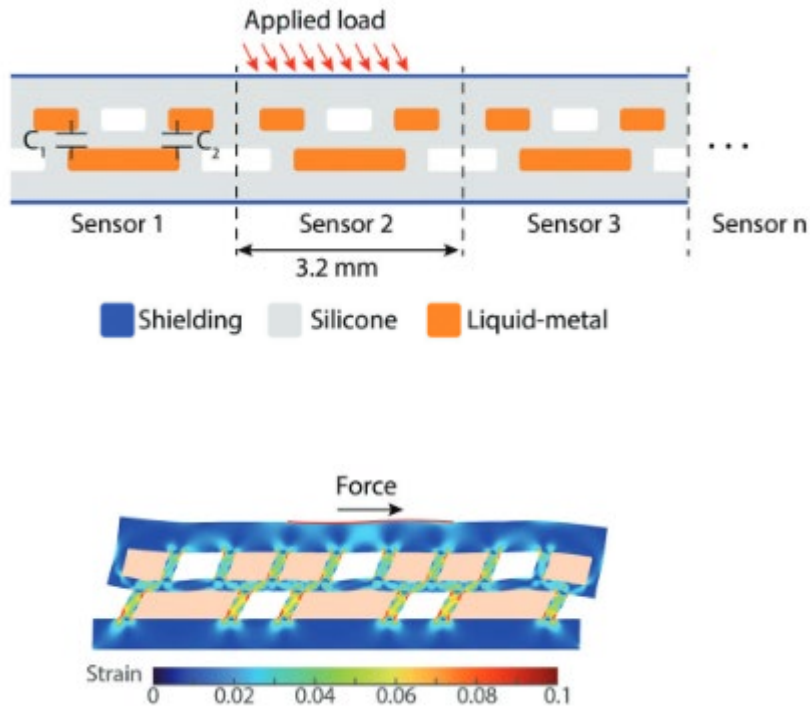
Yes, we can!



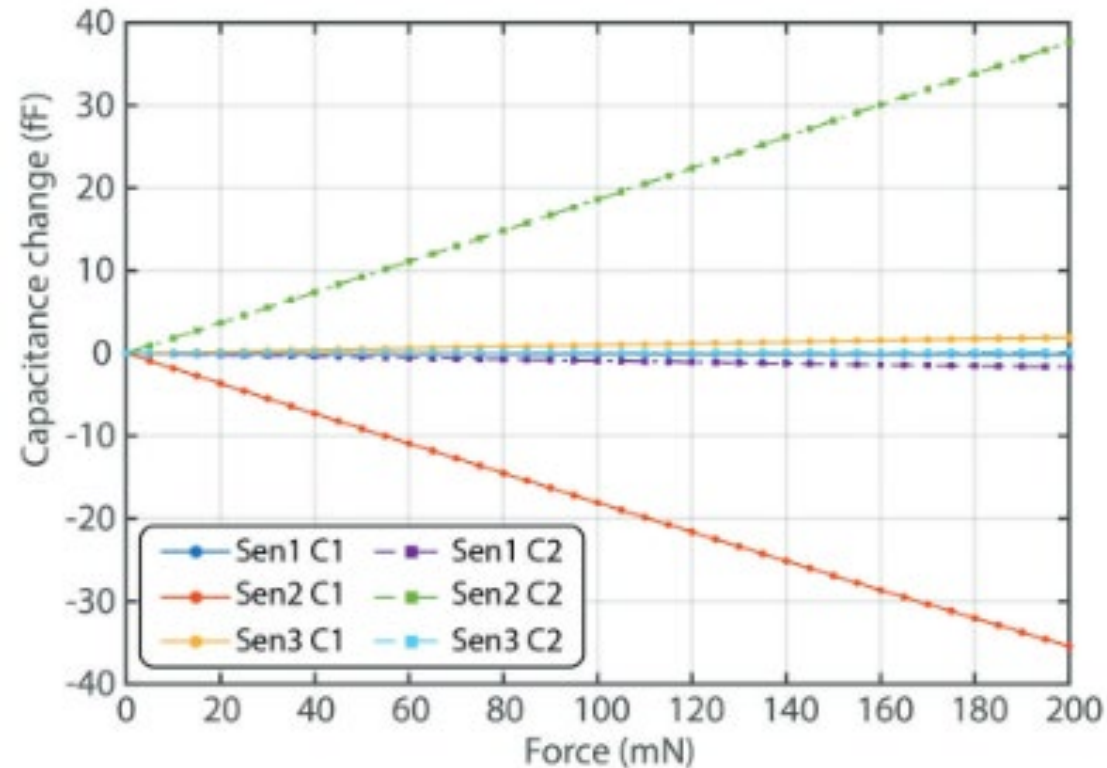
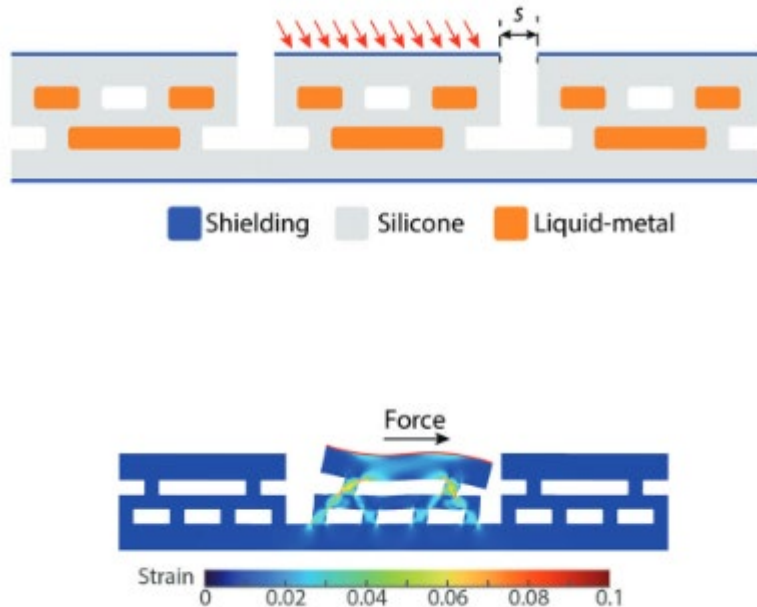
Requirements for an array

- Reasonable spatial resolution (~ 3 mm in our case)
- Each unit reporting data independent of its neighbours (no crosstalk)
- Easy to connect to each unit

Simply attaching multiple sensors together doesn't work



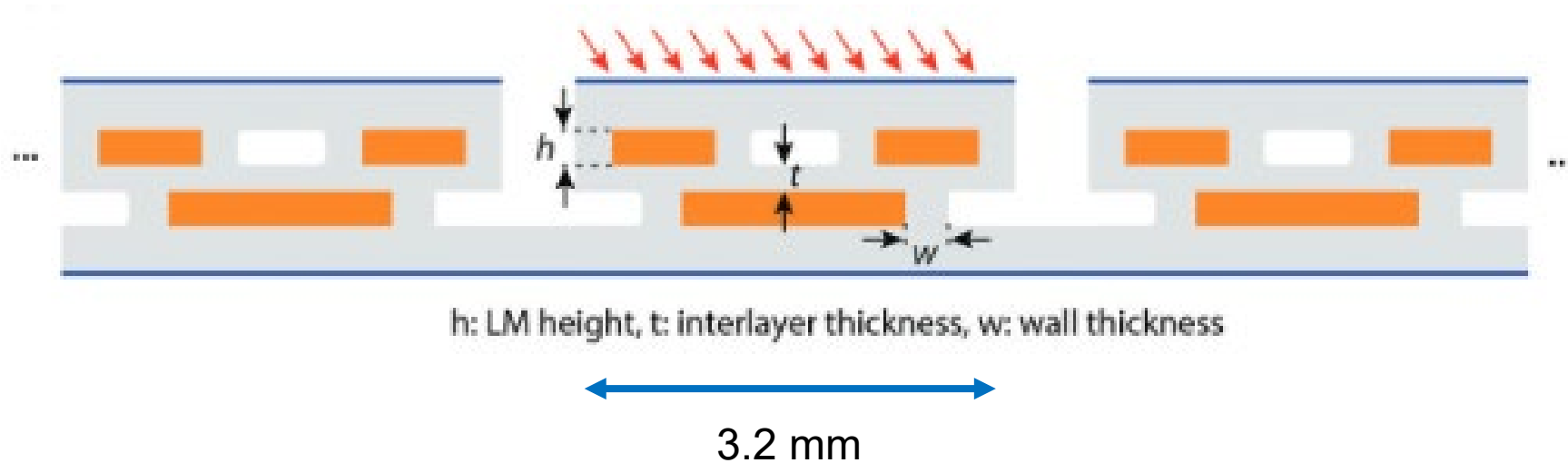
The units must be separated if force location is to be determined



There is still crosstalk if the gap is insufficient

We repeat this with increasing spacing and determined 2 mm to be a suitable value

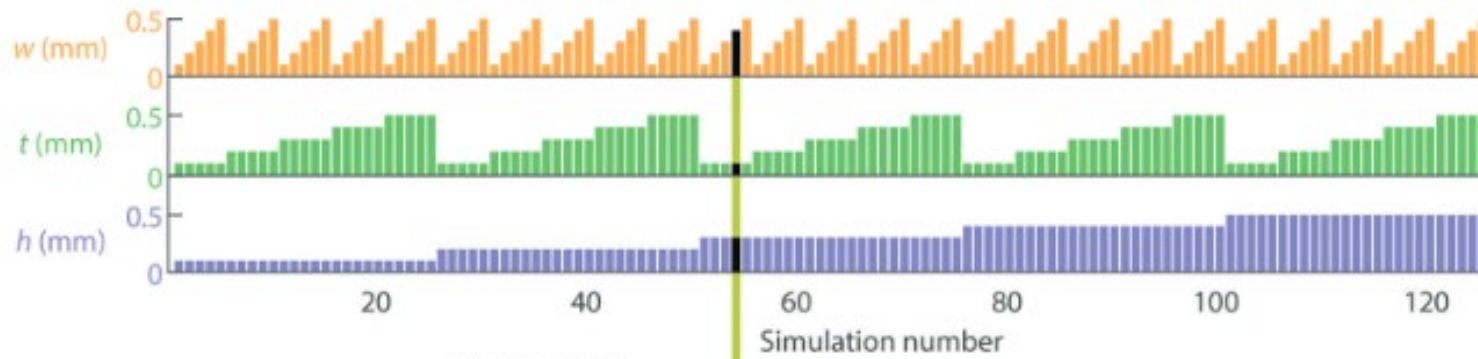
What about the dimensions of each unit?



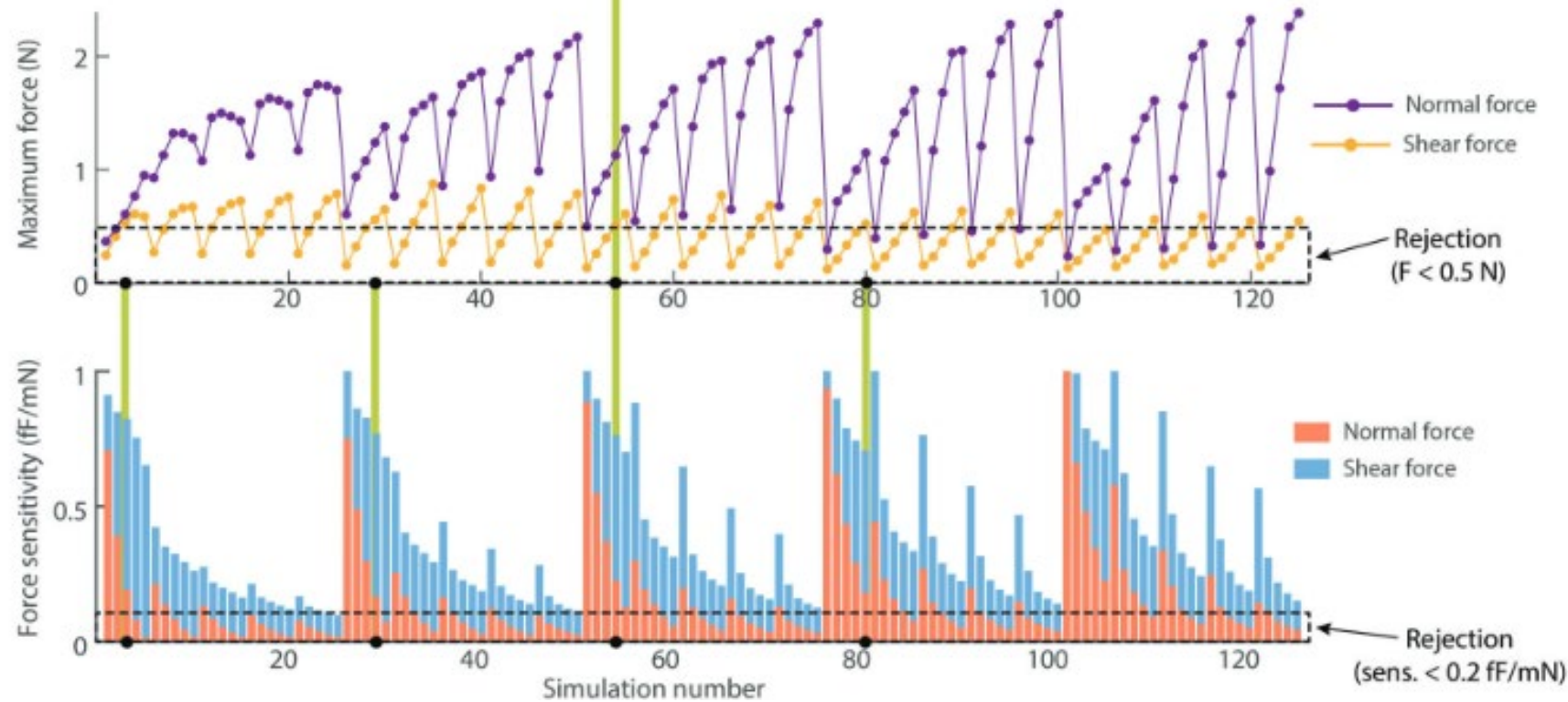
Changing h , t and w changes the load bearing capacity and sensitivity of the sensor.

Max force

Change in capacitance
per unit change in force



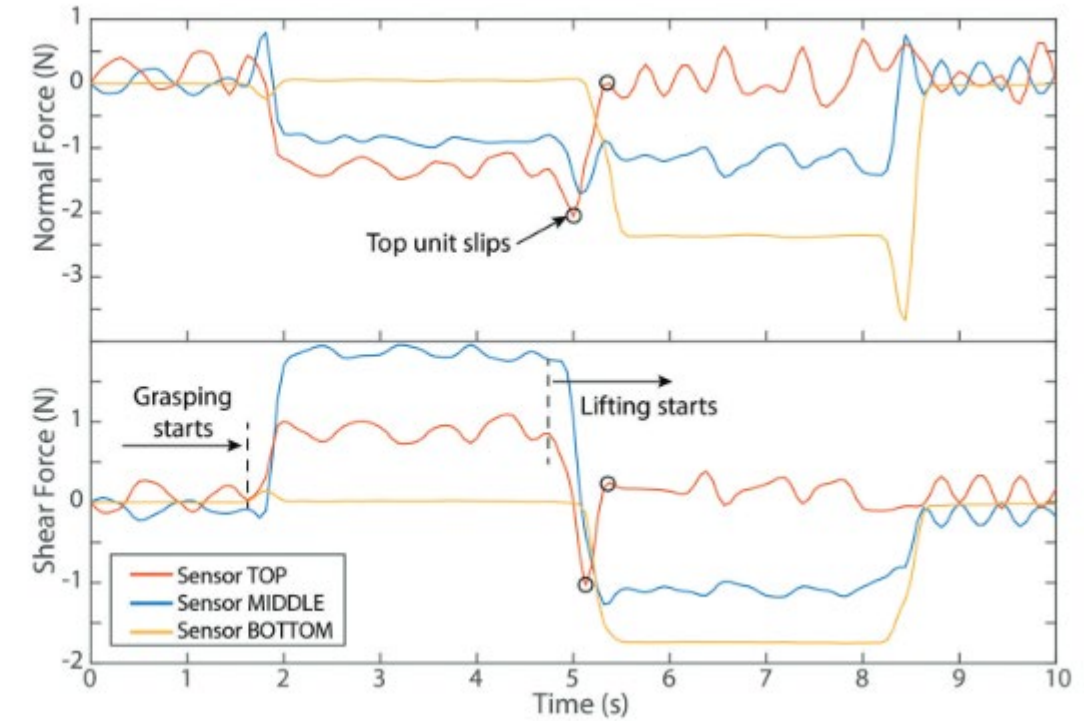
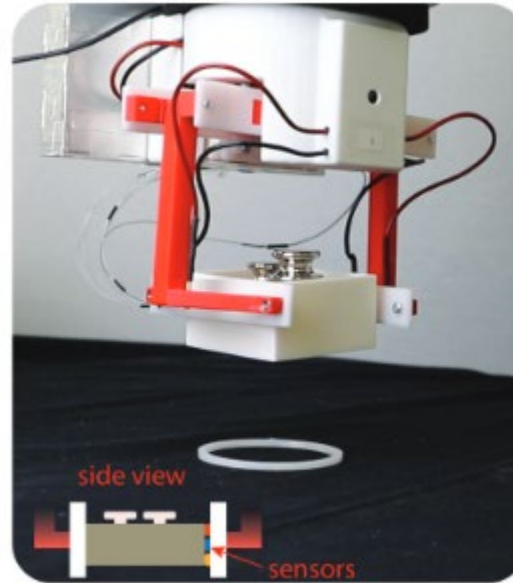
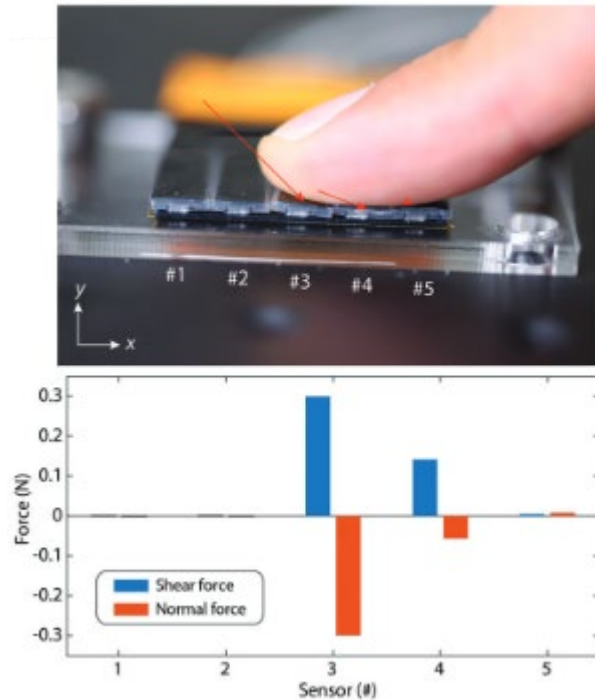
Chosen design:
 $w = 400$ microns
 $h = 300$ microns
 $t = 100$ microns



Below this is too weak

We can't detect below this

We used the sensor for local slip detection



They are good... but can be better!

- How about multi-axis force-torque sensing?
[Low profile 3D sensors](#)
- Liquid metal is not easy to work with.
[Alternative materials and rapid manufacturing](#)
- Moulding the air pockets and channels has a success rate of about 20%.
[New ways to tune stiffness](#)



Genevieve Hughes



Noon Osman

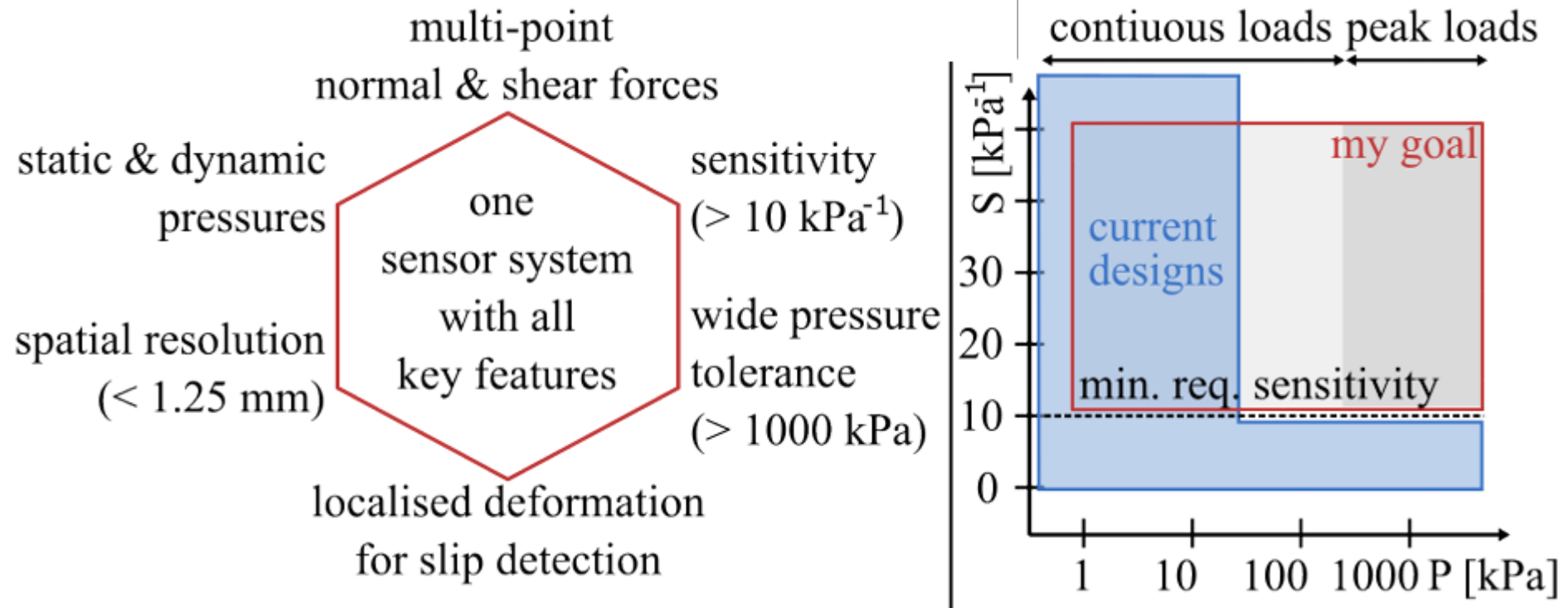


Huu Khang Nguyen



Samuel Di Bartolo

What I am working towards



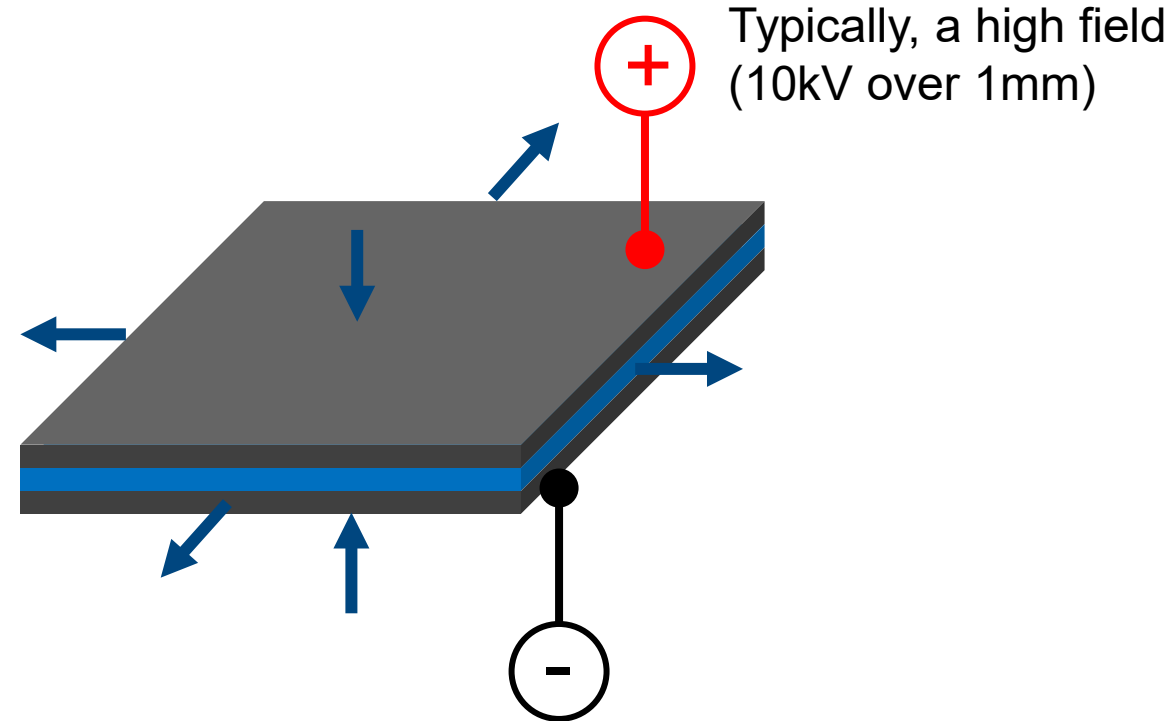
Electrostatically induced actuation

Dielectric elastomer actuators

Efficiency > 80%

Compliant electrodes

Insulating, elastic membrane

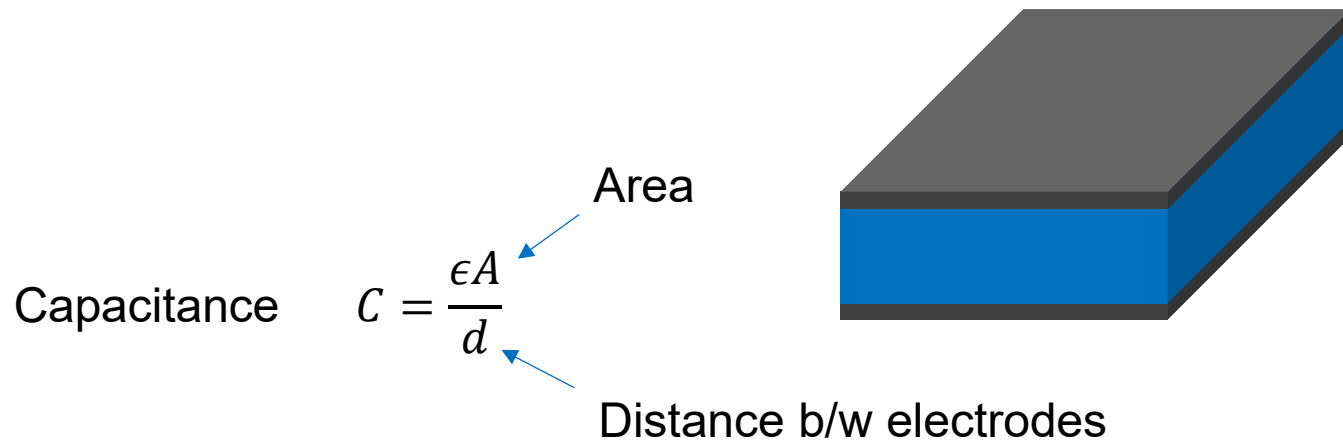


Can we sense the state of actuation?

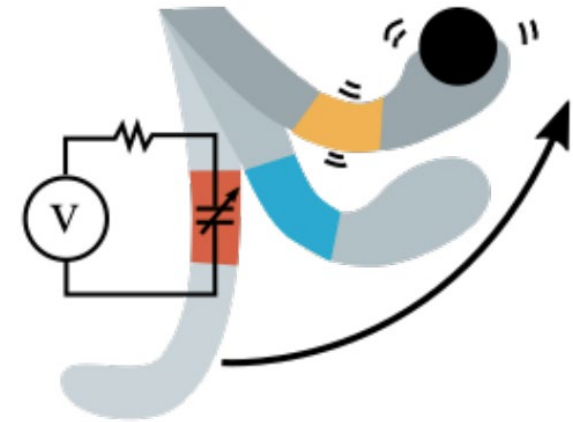
Can we sense the state of actuation?

Notice that the layers form a capacitor

We get a sensor for free!



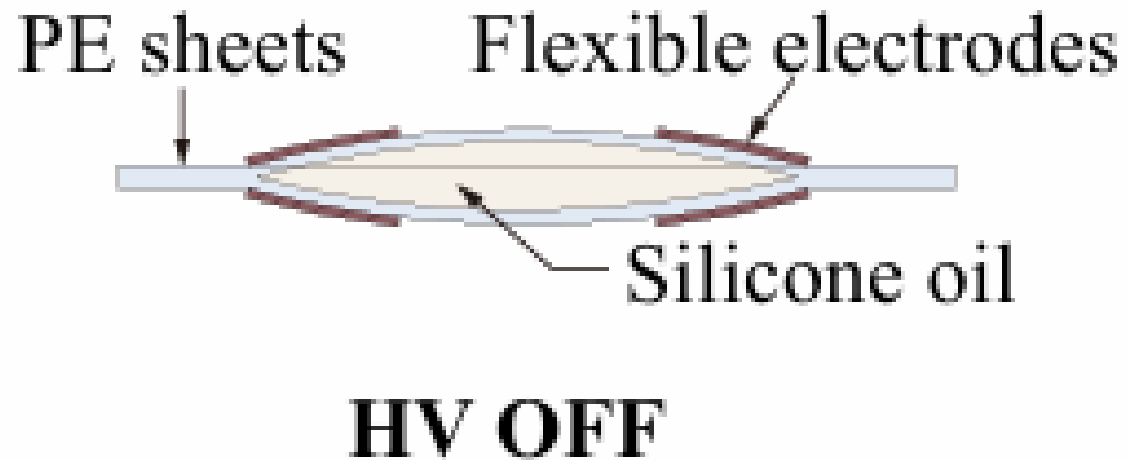
Area and distance depend on the deformation of the structure.



So, closed loop control is possible.

Can we feel what the sensor is measuring?

Electro-hydraulic actuators



Zapping polymers with electric fields is useful



Stiffness change



Adhesion



Sensing



Actuation



Haptics

Soft Robotics Australia

2nd Soft Robotics Symposium

8th December at Deakin's Downtown Campus in Melbourne

<https://www.softrobotics.au/home>



Thank you

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QUT Centre for Robotics